

DEPI – Digital Egypt Pioneers Initiative

Data Science & Artificial Intelligence Track

Requirements Gathering

Requirements Reports

Stakeholder Analysis | Requirements | User Stories & Use Cases

SMEs Risk Prediction — Machine Learning Model

Machine Learning & AI Team | DEPI 2026

1. Stakeholder Analysis

1.1 Stakeholder Segmentation

Group	Stakeholders Included
- Business & RiskManagement	Bank Management, Risk Management Department, Executive Leadership, Finance Team.
- Credit & Loan Operations	Loan Officers, Credit Officers, Risk Analysts.
-Regulatory & Compliance	Regulatory Bodies, Compliance Authorities.
-Technical Team	Data Scientists, Data Science Team, IT Team, System Administrators.
- SME Clients	SME Owners, SME Clients.
-Project Development Team	Project Team, Developers.

1.2 Detailed Stakeholder Analysis

Stakeholder Group	Role	Primary Goal	Key Needs
Business & Risk Management	Strategic decision-making, risk policy definition	Reduce loan default rates, Maximize ROI	High recall, explainable decisions, alignment with business KPIs.
Credit & Loan Operations	Daily loan application processing, manual review of borderline cases	Fast and reliable risk classification	Simple interface, quick predictions, probability scores for borderline cases.
Regulatory & Compliance	Ensure fairness, prevent discrimination, audit model decisions	Compliance with Basel III and fair lending laws	Model transparency, audit trails, bias detection reports.
Technical Team	Model development, deployment, maintenance, and monitoring	Build accurate, scalable, and maintainable system	Clean data, reproducible experiments, saved model (.pkl), performance metrics.
SME Clients (End Users)	Apply for financing through the bank	Fair and fast treatment, clear decision reasons	Transparency, quick response, explanation for High Risk classification.
Project Development Team	Build the system from scratch according to requirements	Deliver on time, meet all specifications	Clear requirements, proper tools, complete documentation.

2. Functional Requirements (FR)

- **FR-01** – The system shall allow loan officers to input SME financial and business data (e.g., income, credit score, debt ratio).
- **FR-02** – The system shall preprocess input data (handling missing values, scaling, encoding).
- **FR-03** – The system shall predict the risk level of an SME (e.g., Low Risk / High Risk).
- **FR-04** – The system shall generate a risk score (probability of default).

- **FR-05** – The system shall display prediction results to the user in a clear format.
- **FR-06** – The system shall provide an explanation for each prediction (feature importance or reason).
- **FR-07** – The system shall flag borderline cases for manual review by risk analysts.
- **FR-08** – The system shall store prediction results and input data for future reference.
- **FR-09** – The system shall allow authenticated access with role-based permissions.
- **FR-10** – The system shall load SME dataset from CSV file (minimum 4000 rows, minimum 10 features).
- **FR-11** – The system shall create a binary target column risk_sharp using configurable financial rules.
- **FR-12** – The system shall remove unnecessary columns (IDs, dates, free text) automatically.
- **FR-13** – The system shall perform one-hot encoding for all categorical variables.
- **FR-14** – The system shall scale numerical features using StandardScaler.
- **FR-15** – The system shall split data into training (80%) and testing (20%) sets with stratification.
- **FR-16** – The system shall train at least: Logistic Regression and Random Forest classifiers.
- **FR-17** – The system shall compute and display: Accuracy, Precision, Recall, F1-Score, ROC-AUC.
- **FR-18** – The system shall generate and display Confusion Matrix.
- **FR-19** – The system shall plot ROC curve for visual evaluation.
- **FR-20** – The system shall calculate and display Feature Importance (for Random Forest).
- **FR-21** – The system shall save the best performing model as .pkl file.
- **FR-22** – The system shall save the scaler as .pkl file for future predictions.
- **FR-23** – The system shall generate a clean CSV containing features + target column.
- **FR-24** – The system shall allow prediction for a new single SME using saved model and scaler.

3. Non-Functional Requirements (NFR)

Performance

- **NFR-P01** – The system shall generate predictions in less than 2 seconds per request.
 - **NFR-P02** – The system shall support multiple concurrent users without performance degradation.
 - **NFR-P03** – Model training time for 5000 records shall not exceed 5 minutes.
 - **NFR-P04** – Batch prediction for 1000 SMEs shall complete within 30 seconds.
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Accuracy

- **NFR-A01** – The model shall achieve Recall $\geq 85\%$ for detecting high-risk SMEs.
 - **NFR-A02** – The model shall achieve ROC-AUC ≥ 0.90 .
 - **NFR-A03** – The model shall achieve Precision $\geq 80\%$ for the high-risk class.
 - **NFR-A04** – The model shall achieve overall Accuracy $\geq 85\%$ on test data.
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Usability

- **NFR-U01** – The system shall provide a simple and user-friendly interface for loan officers.
 - **NFR-U02** – The system shall display results in a clear and understandable format.
 - **NFR-U03** – All code shall be written in Jupyter Notebook with clear comments.
 - **NFR-U04** – Outputs shall be printed with clear labels (metrics, confusion matrix, etc.).
 - **NFR-U05** – Error messages shall be descriptive and actionable.
 - **NFR-U06** – The system shall be runnable by someone with basic Python knowledge.
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Explainability

- **NFR-E01** – The system shall provide explanations for each prediction (feature importance or specific reason).
 - **NFR-E02** – The system shall display which features most influenced a High Risk decision.
 - **NFR-E03** – The system shall allow users to see the probability score behind each prediction.
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Security

- **NFR-S01** – The system shall ensure data privacy and protect sensitive SME information.
 - **NFR-S02** – The system shall implement role-based access control (RBAC).
 - **NFR-S03** – No hard-coded credentials or API keys shall exist in the code.
 - **NFR-S04** – Input validation shall be performed on all data fields.
 - **NFR-S05** – The system shall not execute arbitrary code from CSV inputs.
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Reliability

- **NFR-R01** – The system shall maintain uptime of at least 99.5%.
 - **NFR-R02** – The system shall log all system activities for auditing and recovery.
 - **NFR-R03** – No missing values shall exist in the final processed dataset.
 - **NFR-R04** – Model training shall produce consistent results (fixed random_state).
 - **NFR-R05** – The system shall handle empty CSV files gracefully with an error message.
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Maintainability

- **NFR-M01** – The system shall allow easy model retraining with new data.
- **NFR-M02** – The system shall store the trained model (e.g., .pkl file) for reuse.

- **NFR-M03** – Code shall be modular with logical cell separation in the notebook.
 - **NFR-M04** – Target construction logic shall be clearly documented.
 - **NFR-M05** – Feature selection process shall be documented.
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Scalability

- **NFR-SC01** – The system shall scale to handle increasing numbers of SME applications.
 - **NFR-SC02** – The system shall handle up to 50,000 rows without crashing.
 - **NFR-SC03** – The model shall be retrainable with new data without code changes.
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Compliance

- **NFR-C01** – The system shall comply with financial regulations and fairness requirements.
- **NFR-C02** – No bias shall be detected across different SME segments (industry, region, size).
- **NFR-C03** – The system shall provide audit trails for compliance review.
- **NFR-C04** – The system shall generate fairness reports upon request.

4. User Stories & Use Cases

4.1 User Stories

- **US-01:** As a loan officer, I want to enter SME data, so that I can evaluate the loan application.
- **US-02:** As a loan officer, I want to receive a risk prediction, so that I can make a quick decision.
- **US-03:** As a loan officer, I want to upload a CSV file with multiple SME records, so that I can process batch applications.
- **US-04:** As a loan officer, I want to see the probability score behind each prediction, so that I know how confident the model is.
- **US-05:** As a risk analyst, I want to review borderline cases, so that I can make more accurate decisions.
- **US-06:** As a risk analyst, I want to understand why the model flagged a case as high risk, so that I can trust the system.
- **US-07:** As a risk analyst, I want to see which features influenced a High Risk decision, so that I can explain to clients.
- **US-08:** As a risk manager, I want to adjust the risk threshold, so that I can balance between catching risks and avoiding false alarms.
- **US-09:** As a risk manager, I want to generate performance reports, so that I can monitor model effectiveness over time.
- **US-10:** As an SME client, I want to receive a decision quickly, so that I know whether I will get the loan.
- **US-11:** As an SME client, I want to understand the reason for rejection, so that I can improve my application.
- **US-12:** As a compliance officer, I want to review model fairness, so that I can ensure no bias exists.

- **US-13:** As a compliance officer, I want to access audit logs, so that I can trace decisions for regulatory review.
- **US-14:** As a data scientist, I want to retrain the model with new data, so that I can improve prediction accuracy.
- **US-15:** As a data scientist, I want to save the trained model and scaler, so that I can deploy without retraining every time.
- **US-16:** As a system administrator, I want to monitor system uptime and performance, so that I can ensure system stability.
- **US-17:** As a system administrator, I want to access system logs, so that I can diagnose and fix issues quickly.
- **US-18:** As a bank manager, I want to view overall model performance metrics, so that I can make strategic decisions.
- **US-19:** As a bank manager, I want to see reduction in default rates over time, so that I can justify continued investment.

4.2 Use Case Scenarios

Use Case #UC-01: Predict SME Risk (Single)

Actors: Loan Officer

Description: Loan officer enters financial data for one SME and receives an immediate risk prediction

Preconditions: Loan officer is authenticated and has access to SME financial data

Main flow:

1. The loan officer logs into the system with authorized credentials
2. The loan officer opens the single prediction form from the main screen
3. The loan officer enters the required SME financial data
4. The system validates all inputs are within acceptable ranges
5. The system preprocesses the data (scaling and encoding)
6. The system loads the trained Random Forest model
7. The system generates risk prediction (High Risk / Low Risk)
8. The system displays the risk label and probability score
9. The system logs the prediction for audit purposes

Alternative flows:

- The loan officer enters missing required field → The system displays error message identifying the missing field
- The loan officer enters value out of acceptable range → The system displays error with acceptable range
- The model file is not found → The system displays error and alerts system administrator

Dependencies: Trained model file (final_risk_model.pkl) and scaler file (final_scaler.pkl) must be available

Use Case #UC-02: Predict SME Risk (Batch)

Actors: Loan Officer / Risk Analyst

Description: User uploads a CSV file with multiple SME records and receives predictions for all

Preconditions: User is authenticated and has CSV file with correct column format

Main flow:

1. The user logs into the system with authorized credentials
2. The user opens the batch prediction upload screen
3. The user clicks "Upload CSV" button
4. The user selects CSV file from local storage
5. The system validates file format and required columns
6. The system processes each record through the trained model
7. The system generates risk predictions for all records
8. The system displays summary metrics (total predictions, risk distribution)
9. The system provides option to download results as CSV

Alternative flows:

- The user uploads invalid file format → The system rejects upload with error message
- The CSV file has missing required columns → The system lists missing columns and stops processing
- The CSV file is empty → The system displays error message

Dependencies: Trained model file (final_risk_model.pkl) must be available

Use Case #UC-03: Review Borderline Case

Actors: Risk Analyst

Description: System automatically flags predictions with probability close to 0.5 for manual review

Preconditions: Batch or single prediction has been completed

Main flow:

1. The risk analyst logs into the system with authorized credentials
2. The system automatically identifies borderline cases (probability between 0.4 and 0.6)
3. The system routes flagged cases to Risk Analyst queue
4. The risk analyst opens the borderline cases review dashboard
5. The risk analyst reviews each flagged application details
6. The risk analyst examines feature importance for each flagged case
7. The risk analyst makes final decision (override model or accept)
8. The system records override decision in audit log

Alternative flows:

- No borderline cases exist → The system displays "No cases pending review" message
- Risk analyst overrides model decision → The system requires reason for override before saving

Dependencies: Risk probability scores must be generated by the model

Use Case #UC-04: View Decision Explanation (Client)

Actors: SME Client (via Loan Officer)

Description: Client receives explanation for a High Risk prediction

Preconditions: A risk prediction has been made and client requests explanation

Main flow:

1. The SME client submits application through the bank
2. The system processes the request through the risk model
3. The system generates risk prediction

4. The system prepares explanation for High Risk prediction
5. The loan officer receives the decision
6. The loan officer communicates decision to SME client
7. If High Risk, the system shows explanation (e.g., "Credit score below 550")
8. The SME client receives reason for rejection

Alternative flows:

- Prediction is Low Risk → The system approves application without detailed explanation
- Client requests more details → Loan officer requests feature importance report

Dependencies: Model must capture feature importance for each prediction

Use Case #UC-05: Retrain Model

Actors: Data Scientist

Description: Data scientist retrains the risk model using updated or expanded dataset

Preconditions: New or updated dataset is available as CSV file

Main flow:

1. The data scientist logs into the development environment
2. The data scientist uploads the new dataset
3. The system preprocesses the data (encoding and scaling)
4. The system splits data into training (80%) and testing (20%)
5. The system trains Logistic Regression and Random Forest models
6. The system evaluates both models (ROC-AUC, Precision, Recall)
7. The system compares model performance
8. The system saves the best performing model as final_risk_model.pkl
9. The system saves the scaler as final_scaler.pkl
10. The system generates performance report for comparison

Alternative flows:

- New dataset has different columns than expected → The system alerts user and stops
- Model performance decreases compared to previous version → The system warns user before saving

Dependencies: Training script must be configured correctly

Use Case #UC-06: Compliance Audit

Actors: Compliance Officer

Description: Compliance officer reviews model fairness and regulatory compliance

Preconditions: Model has been trained and tested

Main flow:

1. The compliance officer logs into the system with authorized credentials
2. The compliance officer requests model performance report
3. The system generates classification report by SME segment (industry, region, size)
4. The system displays feature importance analysis
5. The system shows confusion matrix with bias detection metrics
6. The system exports audit log of all predictions
7. The compliance officer reviews for any signs of unfair discrimination
8. The compliance officer approves model or flags issues

Alternative flows:

- Bias detected across any segment → The compliance officer flags model for retraining
- Audit logs are missing → The system administrator investigates logging issue

Dependencies: Audit logs must be maintained for all predictions

Use Case #UC-07: System Monitoring

Actors: System Administrator

Description: Administrator monitors system health and performance

Preconditions: System is deployed and running

Main flow:

1. The system administrator logs into the monitoring dashboard
2. The system displays uptime status (target $\geq 99.5\%$)
3. The system shows response time metrics for predictions
4. The system displays total number of predictions made (daily/weekly/monthly)
5. The system shows any errors or warnings from the past 24 hours
6. The system administrator reviews system logs for issues
7. The system administrator takes corrective action if needed

Alternative flows:

- Uptime falls below 99.5% → The system sends alert to administrator
- Response time exceeds 2 seconds → The system logs performance issue for review

Dependencies: System must have logging and monitoring capabilities

Use Case #UC-08: Generate Management Report

Actors: Bank Manager

Description: Manager views overall model performance and business impact

Preconditions: Model has been in use for a period of time

Main flow:

1. The bank manager logs into the system with authorized credentials
2. The bank manager opens the reports dashboard
3. The bank manager selects report period (monthly/quarterly)
4. The system generates summary of model metrics (Accuracy, Recall, Precision)
5. The system shows trend of default rates over time
6. The system displays total number of applications processed
7. The system shows percentage of automated decisions vs manual reviews
8. The bank manager reviews report for strategic decision-making

Alternative flows:

- No data available for selected period → The system displays message requesting wider time range
- Metrics show significant degradation → The system recommends model retraining

Dependencies: Performance metrics must be stored historically